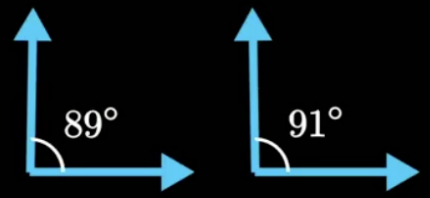
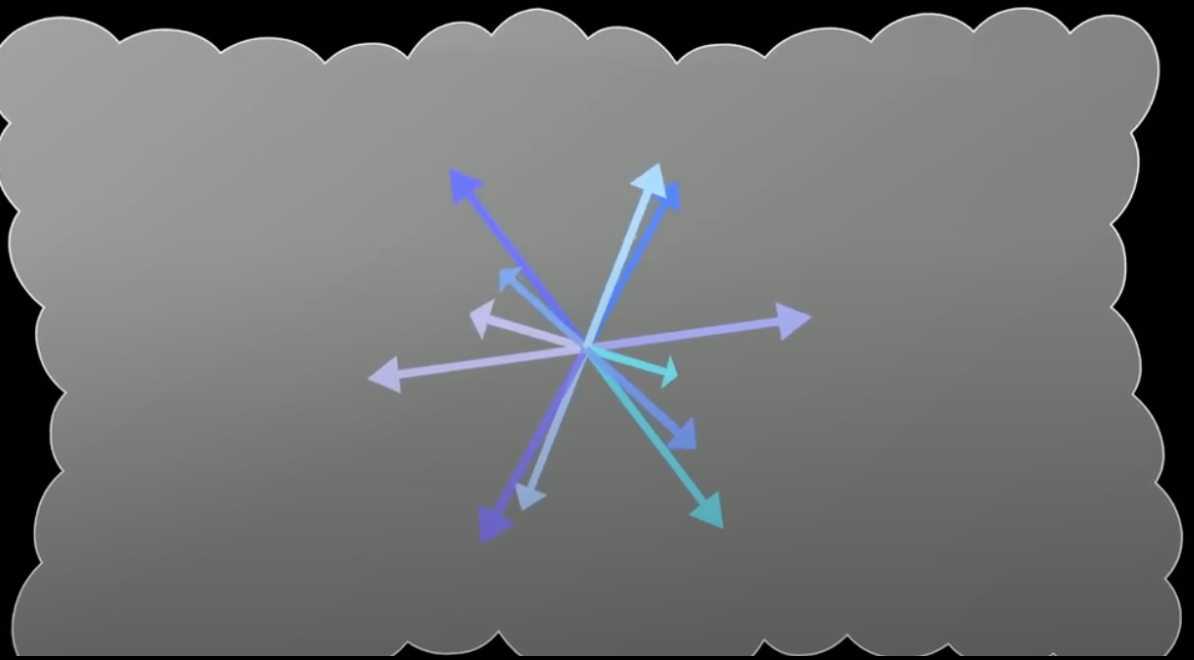
The lemma likely explains how large language models (LLMs) like transformers are able to represent highly nuanced word meanings and context in relatively low-dimension embeddings.[2]



Choose multiple vectors,
each pair between 89° and 91° apart

Johnson–Lindenstrauss Lemma $\Rightarrow$ Maximum # of vectors: $\approx \exp(\epsilon \cdot N)$

N-dimensional Space



The **Johnson–Lindenstrauss (JL) Lemma** is a foundational result in high-dimensional geometry that explains *how complex, high-dimensional relationships can be preserved under surprisingly strong dimensionality reduction*.

🧭 The Lemma Itself

Statement (informal):
Given any set of n points in a high-dimensional space $\mathbb{R}^d$, and any small error tolerance $\varepsilon \in (0,1)$, there exists a linear map
$f : \mathbb{R}^d \to \mathbb{R}^k$
with $k = O\left(\frac{\log n}{\varepsilon^2}\right)$
such that for all pairs of points x_i, x_j,
$(1-\varepsilon)\|x_i - x_j\|^2 \leq \|f(x_i) - f(x_j)\|^2 \leq (1+\varepsilon)\|x_i - x_j\|^2.$

In other words:
you can randomly project very high-dimensional data into a space of dimension only logarithmic in the number of samples — *without distorting pairwise distances by more than a small factor*.

🧠 How It Relates to LLMs and Word Embeddings

LLMs (Large Language Models) rely on *embeddings* — vector representations of tokens or words in high-dimensional spaces (often 768–8192 D).
The **semantic relationships** among words (similarity, analogy, nuance) are expressed geometrically as distances and directions in this embedding space.

The JL lemma gives a **mathematical explanation** for why these relatively compact embeddings can capture the nuances of meaning in the vast combinatorial space of language:

1. **Preservation of Semantic Structure:**

Even though the "true" semantic space of language may be enormously high-dimensional (reflecting culture, context, syntax, pragmatics, etc.), the lemma assures us that a *random or learned projection* into a much smaller space can *approximately preserve relative distances*.

→ Word pairs like ("king", "queen") and ("man", "woman") retain consistent geometric relationships even in 1–2 k dimensions.

2. **Feasibility of Compression:**

The number of distinct word-meaning relationships that matter in practice (the "n" points) is vastly smaller than the total possible combinations of meanings.

By JL, you only need dimension

$k \sim \frac{\log n}{\varepsilon^2}$.

For example, if $n = 10^6$ meaningful tokens and $\varepsilon = 0.1$,

then $k \approx 10^3$—about the embedding sizes used in GPT-like models.

3. **Robustness of Linear Projections:**

Transformer layers mix representations linearly (plus nonlinear gating).

Because linear projections approximately preserve geometry (as JL guarantees for random ones), the model can safely compress and recombine information across layers without catastrophic loss of meaning.

4. **Distributed Semantics:**

The lemma supports the intuition that *semantic meaning doesn't need explicit symbolic coordinates*.

Nuances can be represented as *distributed directions* in a continuous space — distances and angles encode relationships robustly under random or learned linear transformations.
